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Turbine shroud assemblies with strip seals

Abstract

A turbine shroud assembly comprising a first shroud segment, a second shroud segment, and a plurality of seals. The first shroud segment includes a first carrier segment and a first blade track segment having a first shroud wall that is formed to include a first recess that extends circumferentially into the first shroud wall. The second shroud segment includes a second carrier segment and a second blade track having a second shroud wall that is formed to include a second recess that extends circumferentially into the second shroud wall. The plurality of seals extend circumferentially into the first shroud segment and the second shroud segment to block gases from escaping the gas path radially between the first shroud segment and the second shroud segment.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to turbine shroud assemblies, and more specifically to sealing of turbine shroud assemblies used with gas turbine engines.

BACKGROUND

(2) Gas turbine engines are used to power aircraft, watercraft, power generators, and the like. Gas turbine engines typically include a compressor, a combustor, and a turbine. The compressor compresses air drawn into the engine and delivers high pressure air to the combustor. In the combustor, fuel is mixed with the high pressure air and is ignited. Products of the combustion

reaction in the combustor are directed into the turbine where work is extracted to drive the compressor and, sometimes, an output shaft. Left-over products of the combustion are exhausted out of the turbine and may provide thrust in some applications.

(3) Compressors and turbines typically include alternating stages of static vane assemblies and rotating wheel assemblies. The rotating wheel assemblies include disks carrying blades around their outer edges. When the rotating wheel assemblies turn, tips of the blades move along blade tracks included in static shrouds that are arranged around the rotating wheel assemblies. Such static shrouds may be coupled to an engine case that surrounds the compressor, the combustor, and the turbine.

(4) Some shrouds are made up of a number of segments arranged circumferentially adjacent to one another to form a ring. Such shrouds may include sealing elements between segments to block air from leaking through the segments during operation of the gas turbine engine.

SUMMARY

(5) The present disclosure may comprise one or more of the following features and combinations thereof.

(6) A turbine shroud assembly for use with a gas turbine engine may comprise a first shroud segment, a second shroud segment, and a plurality of seals. The first shroud segment may include a first carrier segment arranged circumferentially at least partway around a central axis and a first blade track segment supported by the first carrier segment to define a first portion of a gas path of the turbine shroud assembly. The first blade track segment may have a first shroud wall, a first attachment flange, and a second attachment flange. The first shroud wall may extend circumferentially partway around the central axis. The first attachment flange may extend radially outward from the first shroud wall. The second attachment flange may extend radially outward from the first shroud wall axially spaced apart from the first attachment flange. The first shroud wall may have a first radial outer surface and a first radial inner surface. The first radial outer surface may include a first portion and a second portion that extends circumferentially away from the first portion and is spaced radially outward from the first portion. The first shroud wall may be formed to include a first recess that extends circumferentially into the first shroud wall.

(7) In some embodiments, the second shroud segment may be arranged circumferentially adjacent the first shroud segment about the central axis. The second shroud segment may include a second carrier segment arranged circumferentially at least partway around the central axis and a second blade track segment supported by the second carrier segment to define a second portion of the gas path of the turbine shroud assembly. The second blade track segment may have a second shroud wall, a first attachment flange, and a second attachment flange. The second shroud wall may extend circumferentially partway around the central axis. The first attachment flange may extend radially outward from the second shroud wall. The second attachment flange may extend radially outward from the second shroud wall axially spaced apart from the first attachment flange of the second blade track segment. The second shroud wall may have a second radial outer surface and a second radial inner surface. The second radial outer surface may include a first portion and a second portion that extends circumferentially away from the first portion of the second radial outer surface and is spaced radially outward from the first portion of the second radial outer surface. The second shroud wall may be formed to include a second recess that extends circumferentially into the second shroud wall.

(8) In some embodiments, the plurality of seals may extend circumferentially into the first shroud segment and the second shroud segment to block gases from escaping the gas path radially between the first shroud segment and the second shroud segment. The plurality of seals may include a first strip seal and a second strip seal. The first strip seal may extend axially along the first portion of the first radial outer surface of the first shroud wall and the first portion of the second radial outer surface of the second shroud wall to block the gases from passing radially between and beyond the first shroud wall and the second shroud wall. The second strip seal may be located radially inward

of the first strip seal and may extend circumferentially into the first recess formed in the first shroud wall of the first blade track segment and the second recess formed in the second shroud wall of the second blade track segment to provide a heat shield for the first strip seal to protect the first strip seal from heat of the gases in the gas path and to reduce a pressure load applied to the first strip seal from the gases in the gas path.

(9) In some embodiments, the second strip seal may include an axial segment that extends axially between a first end and a second end thereof opposite the first end. The second strip seal may further include a forward radial segment coupled with the first end of the axial segment and extending axially forward and radially outward from the first end of the axial segment toward the first strip seal. The second end of the axial segment may terminate in an axial direction to cause the forward radial segment and the axial segment to cooperate with the first strip seal to define a cavity radially between the first strip seal and the axial segment of the second strip seal. The cavity may have an at least partially closed axial forward end defined by the forward radial segment and an open axial aft end that allows a first portion of the gases to flow into the cavity. The first portion of the gases in the cavity may have a relatively lower pressure than a second portion of the gases in the gas path that have a relatively higher pressure so that the second portion of the gases urge the second strip seal radially outward within the first recess and the second recess.

(10) In some embodiments, the axial segment of the second strip seal may have a constant width between the first end and the second end thereof and the forward radial segment of the second strip seal may have a tapered width that decreases as the forward radial segment extends radially outward from the first end of the axial segment toward the first strip seal. The first recess of the first shroud wall may include a radially-extending portion that extends radially inward and axially aft and an axially-extending portion that extends axially aft from the first radially-extending portion. The radially-extending portion may extend radially inward from the first portion of the first radial outer surface of the first shroud wall. The axially-extending portion may be radially spaced apart from the first portion of the first radial outer surface of the first shroud wall.

(11) In some embodiments, the axial segment of the second strip seal may extend circumferentially into the axially-extending portion of the first recess and the forward radial segment of the second strip seal may extend circumferentially into the radially-extending portion of the first recess. The first recess of the first shroud wall may include a radially-extending portion that extends radially inward and axially aft and an axially-extending portion that extends axially aft from the radially-extending portion. The second recess of the second shroud wall may include a radially-extending portion that extends radially inward and axially aft and an axially-extending portion that extends axially aft from the radially-extending portion of the second recess.

(12) In some embodiments, the first carrier segment may include a first outer wall, a first flange that extends radially inward from the first outer wall, and a second flange axially spaced apart from the first flange and extending radially inward from the first outer wall. The first strip seal may include a body segment that extends axially along the first portion of the first radial outer surface of the first shroud wall and the first portion of the second radial outer surface of the second shroud wall, a forward segment coupled to a first end of the body segment and extending axially forward and radially outward from the first end of the body segment into the first flange of the first carrier segment, and an aft segment coupled to a second end of the body segment opposite the first end thereof and extending axially aft and radially outward from the second end of the body segment. The aft segment may be located axially aft of the second flange of the first carrier segment.

(13) In some embodiments, the first strip seal may be formed to include at least one hole extending radially through the first strip seal to direct cooling air radially inward through the at least one hole toward the second strip seal to cool the second strip seal.

(14) According to another aspect of the present disclosure, a turbine shroud assembly for use with a gas turbine engine may comprise a first shroud segment, a second shroud segment, and a plurality of seals. The first shroud segment may include a first carrier segment arranged circumferentially at

least partway around a central axis and a first blade track segment supported by the first carrier segment to define a first portion of a gas path of the turbine shroud assembly. The first blade track segment may have a first shroud wall that extends circumferentially partway around the central axis and a first attachment feature that extends radially outward from the first shroud wall. The first shroud wall may define a first radial outer surface. The second shroud segment may be arranged circumferentially adjacent the first shroud segment about the central axis. The second shroud segment may include a second carrier segment arranged circumferentially at least partway around the central axis and a second blade track segment supported by the second carrier segment to define a second portion of the gas path of the turbine shroud assembly. The second blade track segment may have a second shroud wall that extends circumferentially partway around the central axis and a second attachment feature that extends radially outward from the second shroud wall. The second shroud wall may define a second radial outer surface.

(15) In some embodiments, the plurality of seals may extend circumferentially into the first shroud segment and the second shroud segment. The plurality of seals may include a first seal and a second seal. The first seal may extend axially along the first radial outer surface of the first shroud wall and the second radial outer surface of the second shroud wall. The second seal may be located radially inward of the first seal and may extend circumferentially into the first blade track segment and the second blade track segment.

(16) In some embodiments, the second seal may include an axial segment that extends axially between a first end and a second end thereof opposite the first end and a forward radial segment coupled with the first end of the axial segment and extending axially forward and radially outward from the first end of the axial segment toward the first seal. The second end of the axial segment may terminate in an axial direction to cause the forward radial segment and the axial segment to cooperate with the first seal to define a cavity radially between the first seal and the axial segment of the second seal. The cavity may have an at least partially closed axial forward end defined by the forward radial segment and an open axial aft end.

(17) In some embodiments, the axial segment of the second seal may have a constant width between the first end and the second end thereof and the forward radial segment of the second seal may have a tapered width that decreases as the forward radial segment extends radially outward from the first end of the axial segment toward the first seal. The first shroud wall may be formed to include a first recess that extends circumferentially into the first shroud wall and the second shroud wall may be formed to include a second recess that extends circumferentially into the second shroud wall. The second seal may extend circumferentially into each of the first recess and the second recess to extend therebetween.

(18) In some embodiments, the first carrier segment may include a first outer wall, a first flange that extends radially inward from the first outer wall, and a second flange axially spaced apart from the first flange and extending radially inward from the first outer wall. The first seal may include a body segment, a forward segment, and an aft segment. The body segment may extend axially along the first radial outer surface of the first shroud wall and the second radial outer surface of the second shroud wall. The forward segment may be coupled to a first end of the body segment and may extend axially forward and radially outward from the first end of the body segment into the first flange of the first carrier segment. The aft segment may be coupled to a second end of the body segment opposite the first end thereof and may extend axially aft and radially outward from the second end of the body segment. The aft segment may be located axially aft of the second flange of the first carrier segment.

(19) A method may comprise assembling a first shroud segment by coupling a first blade track segment with a first carrier segment to support the first blade track segment radially inward of the first carrier segment. The method may comprise assembling a second shroud segment by coupling a second blade track segment with a second carrier segment to support the second blade track segment radially inward of the second carrier segment. The method may comprise locating a first

seal on a first radial outer surface of the first blade track segment and a second radial outer surface of the second blade track segment. The method may comprise locating a second seal in a first recess that extends circumferentially into the first blade track segment and in a second recess that extends circumferentially into the second blade track segment so that the second seal is located radially inwardly of the first seal. The method may comprise shielding the first seal from heat of gases in a gas path via the second seal.

(20) In some embodiments, the second seal may include an axial segment that extends axially between a first end and a second end thereof opposite the first end and a forward radial segment coupled with the first end of the axial segment and extending axially forward and radially outward from the first end of the axial segment toward the first seal. The second end of the axial segment may terminate in an axial direction to cause the forward radial segment and the axial segment to cooperate with the first seal to define a cavity radially therebetween. The method may comprise directing a first portion of the gases having a first pressure into the cavity at the second end of the axial segment and urging the second seal radially outward within the first recess and the second recess via a second portion of the gases flowing radially inward of the second seal having a second pressure that is greater than the first pressure.

(21) In some embodiments, the method may comprise urging the first seal radially inward against the first blade track segment and the second blade track segment via a third portion of the gases flowing radially outward of the first seal having a third pressure that is greater than the first pressure.

(22) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cutaway perspective view of a gas turbine engine that includes a fan, a compressor, a combustor, and a turbine, the turbine including a turbine shroud assembly that extends circumferentially around an axis of the gas turbine engine and turbine wheel assemblies that are driven to rotate about the axis to generate power;

(2) FIG. 2 is a cutaway perspective view of a portion of the turbine shroud assembly of FIG. 1 showing one of the turbine wheel assemblies and a first shroud segment of a plurality of shroud segments arranged around the turbine wheel assembly, the first shroud segment including a first carrier segment, a first blade track segment that defines a portion of a gas path of the turbine, and a first retainer that couples the first blade track segment with the first carrier segment, and further showing that a plurality of seals of the turbine shroud assembly extend circumferentially into the first shroud segment to block gases from passing between the first shroud segment and a circumferentially adjacent second shroud segment;

(3) FIG. 3 is a cross-sectional view of the turbine shroud assembly through the plurality of seals of FIG. 2 showing that the first blade track segment includes a first shroud wall that defines a first radial outer surface and is formed to include a first recess that extends circumferentially into the first shroud wall and includes a first radially-extending portion that extends radially inward and axially aft from the first radial outer surface and an axially-extending portion that extends axially aft from the first radially-extending portion, and further showing that a first strip seal of the plurality of seals extends axially along the first radial outer surface of the first shroud wall, a second strip seal of the plurality of seals extends circumferentially into the first recess to provide a heat shield for the first strip seal to protect the first strip seal from heat of gases in the gas path, and a damping segment of the plurality of seals engages the first strip seal to urge the first strip seal radially inwardly;

- (4) FIG. 4 is a cross-sectional view of the first shroud segment of FIG. 3 showing that the first retainer extends through the first blade track segment and through the first carrier segment to couple the first blade track segment to the first carrier segment;
- (5) FIG. 5 is an exploded view of the first and second shroud segments used in the gas turbine engine of FIG. 1 showing the first shroud segment and the second shroud segment spaced apart from the first shroud segment, the second shroud segment including a second carrier segment and a second blade track segment supported by the second carrier segment, and further suggesting that the plurality of seals extends circumferentially into the first shroud segment and the second shroud segment to block gases from escaping the gas path radially between the first shroud segment and the second shroud segment, and further showing that the second strip seal includes an axial segment that extends axially between a first end and a second end thereof and a forward radial segment coupled with the first end of the axial segment to extend radially outward toward the first strip seal;
- (6) FIG. 6 is a cross-sectional diagrammatic view through the first and second shroud segments as assembled in the turbine shroud assembly of FIG. 1 showing that the first shroud segment and the second shroud segment are assembled adjacent one another and the first strip seal and the second strip seal each extend circumferentially between the first shroud segment and the second shroud segment, and further showing that the first radial outer surface of the first shroud wall includes a first portion and a second portion spaced radially outward from the first portion, the first strip seal extends along the first portion of the first radial outer surface of the first shroud wall and a first portion of a second radial outer surface of the second shroud wall and the axial segment of the second strip seal extends into the axially-extending portion of the first recess formed in the first shroud wall of the first blade track segment and into an axially-extending portion of a second recess formed in a second shroud wall of the second blade track segment to protect the first strip seal from heat of the gases in the gas path;
- (7) FIG. 7 is a cross-sectional diagrammatic view through the first and second shroud segments as assembled in the turbine shroud assembly of FIG. 1 showing that the first radially-extending portion of the first recess and a first radially-extending portion of the second recess extend radially outward to the first portion of the first radial outer surface of the first shroud wall and the first portion of the second radial outer surface of the second shroud wall, respectively, and further showing that the forward radial segment of the second strip seal extends radially outward at the first end of the axial segment of the second strip seal toward the first strip seal;
- (8) FIG. 8 is a cross-sectional view of the turbine shroud assembly through the plurality of seals of FIG. 3 showing that the plurality of seals includes the first strip seal and the second strip seal, and further showing that the damping segment is omitted in alternative embodiments;
- (9) FIG. 9 is a cross-sectional view of the turbine shroud assembly through the plurality of seals of FIG. 3 showing that the first strip seal is formed to include at least one hole extending radially therethrough and configured to direct cool air radially inwardly through the at least one hole toward the second strip seal to cool the second strip seal;
- (10) FIG. 10 is a cross-sectional diagrammatic view through the first and second shroud segments as assembled in the turbine shroud assembly of FIG. 9 showing that the first shroud segment and the second shroud segment are assembled adjacent one another and the first strip seal and the second strip seal each extend circumferentially between the first shroud segment and the second shroud segment, and further showing that the first strip seal is formed to include the at least one hole extending radially therethrough to direct cool air toward the second strip seal;
- (11) FIG. 11 is a cross-sectional diagrammatic view of another embodiment of a turbine shroud assembly for use in the gas turbine engine of FIG. 1 showing that a first shroud wall of a first blade track segment is formed to include a first recess and a second shroud wall of a second blade track segment is formed to include a second recess, and a first radially-extending portion of the first recess and a first radially-extending portion of the second recess are tapered toward one another,

and further showing that a second strip seal includes a forward radial segment that extends radially outward toward the first strip seal and is tapered to match a shape of the first and second recesses; (12) FIG. 12 is a cross-sectional view of a turbine shroud assembly for use in the gas turbine engine of FIG. 1 showing an alternative embodiment of a second strip seal, the second strip seal including an axial segment that extends axially between a first end and a second end thereof and an aft radial segment coupled with the second end of the axial segment to extend radially outward and axially aft toward the first strip seal; and

(13) FIG. 13 is a cross-sectional view of a turbine shroud assembly for use in the gas turbine engine of FIG. 1 showing an alternative embodiment of a second strip seal, the second strip seal including an axial segment that extends axially between a first end and a second end thereof, a forward radial segment coupled with the first end of the axial segment to extend radially outward and axially forward toward the first strip seal and an aft radial segment coupled with the second end of the axial segment to extend radially outwardly and axially aft toward the first strip seal.

DETAILED DESCRIPTION

(14) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.

(15) An illustrative aerospace gas turbine engine **10** includes a fan **12**, a compressor **14**, a combustor **16**, and a turbine **18** as shown in FIG. 1. The fan **12** is driven by the turbine **18** and provides thrust for propelling an air vehicle. The compressor **14** compresses and delivers air to the combustor **16**. The combustor **16** mixes fuel with the compressed air received from the compressor **14** and ignites the fuel. The hot, high-pressure products of the combustion reaction in the combustor **16** are directed into the turbine **18** to cause the turbine **18** to rotate about a central axis **11** and drive the compressor **14** and the fan **12**. In some embodiments, the fan **12** may be replaced with a propeller, drive shaft, or other suitable configuration.

(16) The turbine **18** includes at least one turbine wheel assembly **20** and a turbine shroud assembly **22** positioned to surround the turbine wheel assembly **20** as shown in FIGS. 1 and 2. The turbine wheel assembly **20** includes a plurality of blades **21** coupled to a rotor disk **24** for rotation with the rotor disk **24**. The hot, high-pressure combustion products from the combustor **16** are directed toward the blades **21** of the turbine wheel assemblies **20** along a gas path **25**. The turbine wheel assembly **20** further includes a plurality of vanes **15** as shown in FIG. 2. The turbine shroud assembly **22** is coupled to an outer case **17** of the gas turbine engine **10** and extends around the turbine wheel assembly **20** to block gases from passing over the blades **21** during use of the turbine **18** in the gas turbine engine **10**.

(17) The turbine shroud assembly **22** includes a plurality of shroud segments and pluralities of seals between adjacent shroud segments as suggested in FIGS. 2 and 5. Of the plurality of shroud segments, a first shroud segment **26** and a second shroud segment **28** are discussed in detail below. Likewise, a plurality of seals **30** included in the pluralities of seals used in the turbine shroud assembly **22** is shown in FIGS. 2-5. The first shroud segment **26**, the second shroud segment **28**, and the plurality of seals **30** are representative of other adjacent shroud segments and pluralities of seals included in the turbine shroud assembly **22**.

(18) The plurality of seals **30** in the illustrative embodiment includes a first strip seal **44**, a second strip seal **46**, and a damping segment **48** as shown in FIG. 3. The first strip seal **44** blocks gases from the gas path **25** from passing radially between the first shroud segment **26** and the second shroud segment **28**. The second strip seal **46** provides a heat shield for the first strip seal **44** to protect the first strip seal **44** from heat of the gases in the gas path **25**. The second strip seal **46** also reduces a pressure load applied to the first strip seal **44** from the gases in the gas path **25**. The damping segment **48** urges the first strip seal **44** radially inward. In some embodiments, the damping segment **48** is omitted, as shown in FIG. 8. In some embodiments, the plurality of seals **30** includes strip seals **102**, **104**, **106**, **108**, **110** as shown in FIGS. 3 and 5. Any of the strip seals **102**,

104, 106, 108, 110 may be included or omitted from the plurality of seals **30**. The strip seals **102, 104, 106, 108, 110** are representative of more conventional strip seals.

(19) The second shroud segment **28** is arranged circumferentially adjacent the first shroud segment **26** about the central axis **11**. A circumferential gap **G** is formed between the first shroud segment **26** and the second shroud segment **28** as shown in FIG. 6. Though the turbine shroud assembly **22** is shown and described as having two shroud segments **26, 28** and a plurality of seals **30**, the turbine shroud assembly **22** includes additional shroud segments and additional seals so that the turbine shroud assembly **22** extends entirely circumferentially about the central axis **11** as suggested in FIG. 1.

(20) The first shroud segment **26** includes a first carrier segment **32**, a first blade track segment **34**, and a first retainer **36** as shown in FIGS. 3 and 4. The first carrier segment **32** is arranged circumferentially at least partway around the central axis **11** and is coupled with the outer case **17** with hook features in the illustrative embodiment. The first blade track segment **34** is supported by the first carrier segment **32** to define a first portion of the gas path **25**. The first retainer **36** extends axially through the first carrier segment **32** and the first blade track segment **34** to couple the first carrier segment **32** and the first blade track segment **34** together.

(21) The second shroud segment **28** includes a second carrier segment **38**, a second blade track segment **40**, and a second retainer **42** as shown in FIG. 5. The second carrier segment **38** is arranged circumferentially at least partway around the central axis **11** and is coupled with the outer case **17** with hook features in the illustrative embodiment. The second blade track segment **40** is supported by the second carrier segment **38** to define a second portion of the gas path **25**. The second retainer **42** extends axially through the second carrier segment **38** and the second blade track segment **40** to couple the second carrier segment **38** and the second blade track segment **40** together.

(22) The plurality of seals **30** extends circumferentially into the first shroud segment **26** and the second shroud segment **28** as shown in FIG. 6 and as suggested in FIG. 5. The plurality of seals **30**, along with the other strip seals **102, 104, 106, 108, 110**, blocks gases in the gas path **25** from escaping the gas path **25** radially outward and circumferentially between the first shroud segment **26** and the second shroud segment **28** through the circumferential gap **G**.

(23) Degradation and fluttering of strip seals may be a concern in turbine shroud assemblies. To minimize degradation of the first strip seal **44**, the second strip seal **46** is located radially inward of the first strip seal **44** to protect the first strip seal **44** from heat of the gases in the gas path **25**. Further, to minimize fluttering, and thus, reduce the possibility of failure of the first strip seal **44**, the second strip seal **46** reduces a pressure load applied to the first strip seal **44** of the present disclosure from the gases in the gas path **25** so that the first strip seal **44** is urged radially inwardly against the blade track segments **34, 40** and any flutter or vibration is dampened.

(24) Turning back to the first shroud segment **26**, the first carrier segment **32** of the first shroud segment **26** includes a first outer wall **50**, a first flange **52**, and a second flange **54** as shown in FIG. 3. The first flange **52** extends radially inward from the first outer wall **50**. The second flange **54** is axially spaced apart from the first flange **52** and extends radially inward from the first outer wall **50**. The first flange **52** is formed to include a first slot **60** as shown in FIG. 3. The first slot **60** extends circumferentially into the first flange **52** and is shaped to receive a portion of the first strip seal **44** therein.

(25) The first flange **52** of the first carrier segment **32** includes a first wall **62** formed to include a radially inward facing surface **64** as shown in FIG. 3. The first slot **60** extends axially forward and radially outward into the first flange **52** from the radially inward facing surface **64** to match the curvature of a portion of the first strip seal **44** that extends into the first slot **60**. A first protrusion **66** extends radially inward from the first wall **62** axially forward of the first slot **60**. The first protrusion **66** is located axially forward of the first blade track segment **34** to cover an axial forward end **34A** of the first blade track segment **34**. The first protrusion **66** blocks at least a

portion of the gases flowing through the gas path **25** from flowing axially into the first strip seal **44**.
(26) The first carrier segment **32** further includes a third flange **56** and a fourth flange **58** as shown in FIG. **3**. Each of the third flange **56** and the fourth flange **58** extends radially inward from the first outer wall **50**. The third flange **56** is located axially between the first flange **52** and the fourth flange **58**. The fourth flange **58** is located axially between the third flange **56** and the second flange **54**.

(27) The third flange **56** of the first carrier segment **32** is formed to include a second slot **68** as shown in FIG. **3**. The second slot **68** extends radially outward into the third flange **56**. The second slot **68** receives a portion of the damping segment **48** therein as shown in FIG. **3**. The fourth flange **58** of the first carrier segment **32** is formed to include a third slot **70** as shown in FIG. **3**. The third slot **70** extends radially outward into the fourth flange **58**. The third slot **70** receives another portion of the damping segment **48** therein as shown in FIG. **3**. Each of the flanges **52**, **54**, **56**, and **58** of the first carrier segment **32** is formed to include a hole that receives the first retainer **36** therein as shown in FIGS. **3** and **4**. Illustratively, the first carrier segment **32** is made of metallic materials.

(28) The first blade track segment **34** includes a first shroud wall **72** and a first attachment feature **74** that extends radially outward from the first shroud wall **72** as shown in FIGS. **3** and **4**. The first shroud wall **72** extends circumferentially partway around the central axis **11**. The first shroud wall **72** has a first radial outer surface **76** that faces toward the first carrier segment **32** and a first radial inner surface opposite the first radial outer surface **76** that faces toward the gas path **25**.

Illustratively, the first attachment feature **74** includes a first attachment flange **74A** and a second attachment flange **74B** axially aft of the first attachment flange **74A**. Each of the attachment flanges **74A**, **74B** is formed to include a hole that receives the first retainer **36** therein. The first attachment flange **74A** is located axially between the first flange **52** and the third flange **56** as shown in FIG. **3**. The second attachment flange **74B** is located axially between the fourth flange **58** and the second flange **54**. Illustratively, the first blade track segment **34** is made of ceramic matrix composite materials.

(29) The first radial outer surface **76** of the first shroud wall **72** includes a first portion **76A** and a second portion **76B** as shown in FIGS. **5** and **6**. The second portion **76B** is spaced radially outward from the first portion **76A**. The first portion **76A** defines a circumferential end **34B** of the first shroud wall **72** that confronts the second blade track segment **40** as shown in FIG. **6**. The second portion **76B** extends circumferentially away from the first portion **76A**. The circumferential end **34B** is formed with a first pocket **80** that defines the first portion **76A** of the first radial outer surface **76** as shown in FIGS. **5** and **6**. The first shroud wall **72** slopes radially inwardly at the circumferential end **34B** to define the first portion **76A** of the first radial outer surface **76**. The first and second portions **76A**, **76B** of the first radial outer surface **76** are exposed to air located radially between the first carrier segment **32** and the first blade track segment **34**. The first strip seal **44** of the plurality of seals **30** is located on the first portion **76A** of the first radial outer surface **76** as shown in FIG. **6**.

(30) The circumferential end **34B** of the first shroud wall **72** is formed to include a first recess **78** extending circumferentially into the first shroud wall **72** to receive the second strip seal **46** therein as shown in FIG. **6**. The first recess **78** of the first shroud wall **72** includes a first radially-extending portion **78A** that extends radially inward and axially aft and an axially-extending portion **78B** that extends axially aft from the first radially-extending portion **78A** as shown in FIG. **3**. In some embodiments, the first radially-extending portion **78A** extends radially inward from the first portion **76A** of the first radial outer surface **76** of the first shroud wall **72** such that the first radially-extending portion **78A** has an open end at the first portion **76A** as shown in FIG. **7**. The axially-extending portion **78B** is radially spaced apart from the first portion **76A** of the first radial outer surface **76** of the first shroud wall **72**.

(31) In the illustrative embodiment, the first retainer **36** includes a mount pin **37** and a mount plug **39** as shown in FIG. **4**. The first retainer **36** couples the first blade track segment **34** to the first

carrier segment **32** as shown in FIGS. **3** and **4**. The mount pin **37** extends through the first blade track segment **34** and into the first carrier segment **32**. The mount plug **39** fits into the first carrier segment **32** axially aft of the mount pin **37** and circumferentially aligned with the mount pin **37**. In the illustrative embodiment, the mount pin **37** includes a forward pin **41** and an aft pin **43** as shown in FIG. **4**. The forward pin **41** and the aft pin **43** of the mount pin **37** are circumferentially aligned with one another. In this embodiment, the forward pin **41** is separate from the aft pin **43** so as to allow for independent loading during use in the gas turbine engine **10**. In some embodiments, the mount pin **37** is formed as a single pin. Though not shown, in the illustrative embodiment, an additional first retainer is included in the first shroud segment **26** spaced apart circumferentially from the first retainer **36** such that the first shroud segment **26** includes two forward pins **41**, two aft pins **43**, and two mount plugs **39**.

(32) The second carrier segment **38** of the second shroud segment **28** includes a second outer wall **45**, a fifth flange **47**, and a sixth flange **49** as shown in FIG. **5**. The fifth flange **47** extends radially inward from the second outer wall **45**. The sixth flange **49** is axially spaced apart from the fifth flange **47** and extends radially inward from the second outer wall **45**. The fifth flange **47** is formed to include a fourth slot **51** as shown in FIG. **5**. The fourth slot **51** extends into the fifth flange **47** axially forward and radially outward to receive a portion of the first strip seal **44** therein. The first slot **60** and the fourth slot **51** are aligned with one another while the first shroud segment **26** and the second shroud segment **28** are assembled adjacent one another.

(33) The second carrier segment **38** further includes a seventh flange **53** and an eighth flange **55** as shown in FIG. **5**. Each of the seventh flange **53** and the eighth flange **55** extend radially inward from the second outer wall **45**. The seventh flange **53** is located axially between the fifth flange **47** and the eighth flange **55**. The eighth flange **55** is located axially between the seventh flange **53** and the sixth flange **49**. The seventh and eighth flanges **53**, **55** may be inner flanges or clevises that are both located axially inward of the fifth flange **47** and the sixth flange **49**.

(34) The seventh flange **53** of the second carrier segment **38** is formed to include a fifth slot **57** as shown in FIG. **7**. The fifth slot **57** extends radially outward into the seventh flange **53**. The eighth flange **55** of the second carrier segment **38** is formed to include a sixth slot **59** as shown in FIG. **5**. The sixth slot **59** extends radially outward into the eighth flange **55**. The second slot **68** of the first carrier segment **32** and the fifth slot **57** of the second carrier segment **38** are aligned with one another while the first shroud segment **26** and the second shroud segment **28** are assembled adjacent to one another to receive a portion of the damping segment **48** therein. The third slot **70** of the first carrier segment **32** and the sixth slot **59** of the second carrier segment **38** are aligned with one another while the first shroud segment **26** and the second shroud segment **28** are assembled adjacent to one another to receive another portion of the damping segment **48** therein. Each of the flanges **47**, **49**, **53**, and **55** of the second carrier segment **38** is formed to include a hole that receives the second retainer **42** therein.

(35) The second blade track segment **40** includes a second shroud wall **61** and a second attachment feature **63** that extends radially outward from the second shroud wall **61** as shown in FIG. **5**. The second shroud wall **61** has a second radial outer surface **65** that faces toward the second carrier segment **38** and a second radial inner surface opposite the second radial outer surface **65** that faces toward the gas path **25**. The second shroud wall **61** extends circumferentially partway around the central axis **11**. Illustratively, the second attachment feature **63** includes a third attachment flange **63A** and a fourth attachment flange **63B** axially aft of the third attachment flange **63A**. Each of the attachment flanges **63A**, **63B** is formed to include a hole that receives the second retainer **42** therein. The third attachment flange **63A** is located axially between the fifth flange **47** and the seventh flange **53**. The fourth attachment flange **63B** is located axially between the eighth flange **55** and the sixth flange **49**. Illustratively, the second blade track segment **40** is made of ceramic matrix composite materials.

(36) The second radial outer surface **65** of the second shroud wall **61** includes a first portion **65A**

and a second portion **65B** as shown in FIGS. **6** and **7**. The second portion **65B** is spaced radially outward from the first portion **65A**. The first portion **65A** defines a circumferential end **40B** of the second shroud wall **61** that confronts the first blade track segment **34** as shown in FIG. **6**. The second portion **65B** extends circumferentially away from the first portion **65A**. The circumferential end **40B** is formed with a second pocket **69** that defines the first portion **65A** of the second radial outer surface **65** as shown in FIGS. **6** and **7**. The second shroud wall **61** slopes radially inwardly at the circumferential end **40B** to define the first portion **65A** of the second radial outer surface **65**. The first and second portions **65A**, **65B** of the second radial outer surface **65** are exposed to air located radially between the second carrier segment **38** and the second blade track segment **40**. The first strip seal **44** of the plurality of seals **30** is located on the first portion **65A** of the second radial outer surface **65** as shown in FIG. **6**.

(37) The circumferential end **40B** is formed to include a second recess **67** extending circumferentially into the second shroud wall **61** to receive the second strip seal **46** therein as shown in FIG. **6**. The second recess **67** of the second shroud wall **61** includes a first radially-extending portion **67A** that extends radially inward and axially aft and an axially-extending portion **67B** that extends axially aft from the first radially-extending portion **67A**. In some embodiments, the first radially-extending portion **67A** extends radially inward from the first portion **65A** of the second radial outer surface **65** of the second shroud wall **61** such that the first radially-extending portion **67A** has an open end at the first portion **65A** as shown in FIG. **7**. The axially-extending portion **67B** is radially spaced apart from the first portion **65A** of the second radial outer surface **65** of the second shroud wall **61**. The second retainer **42** is the same as the first retainer **36** such that description of the first retainer **36** also applies to the second retainer **42**.

(38) The plurality of seals **30** includes the first strip seal **44**, the second strip seal **46**, and the damping segment **48** shown in FIG. **3**. The first strip seal **44** extends between the first blade track segment **34** and the second blade track segment **40** to block the gases from passing radially between and beyond the first shroud wall **72** and the second shroud wall **61** as shown in FIGS. **6** and **7**. The second strip seal **46** is located radially inward of the first strip seal **44** and extends into the first blade track segment **34** and the second blade track segment **40**. The damping segment **48** is located radially outward of the first strip seal **44** and extends into the first carrier segment **32** and the second carrier segment **38**.

(39) The first strip seal **44** includes a body segment **84**, a forward segment **86**, and an aft segment **88** as shown in FIGS. **3** and **5**. The body segment **84** extends axially along the first portion **76A** of the first radial outer surface **76** of the first shroud wall **72** and the first portion **65A** of the second radial outer surface **65** of the second shroud wall **61** between a first end **84A** and a second end **84B** thereof opposite the first end **84A**. The forward segment **86** is coupled to the first end **84A** of the body segment **84** and extends axially forward and radially outward from the first end **84A** of the body segment **84** into the first slot **60** formed in the first flange **52** of the first carrier segment **32** and the fourth slot **51** formed in the fifth flange **47** of the second carrier segment **38**. Illustratively, at least a portion of the forward segment **86** extends along a curvilinear path and at least another portion of the forward segment **86** extends along a straight path. The forward segment **86** extending into the slots **60**, **51** retains the body segment **84** axially relative to the first shroud segment **26** so that the body segment **84** does not move fore and aft.

(40) The aft segment **88** of the first strip seal **44** is coupled to the second end **84B** of the body segment **84** as shown in FIG. **3**. The aft segment **88** extends axially aft and radially outward from the second end **84B** of the body segment **84**. The aft segment **88** is located axially aft of the second flange **54** of the first carrier segment **32** and the sixth flange **49** of the second carrier segment **38**. In some embodiments, the aft segment **88** abuts an aft wall of the second flange **54** and/or an aft wall of the sixth flange **49**. In some embodiments, the aft segment **88** abuts a vane **15** as suggested in FIG. **3**. Illustratively, at least a portion of the aft segment **88** extends along a curvilinear path and at least another portion of the aft segment **88** extends along a straight path. The engagement of the aft

segment **88** with the second flange **54** axially locates the first strip seal **44** relative to the first shroud segment **26**.

(41) A radial inner surface of the body segment **84** directly contacts the first portions **76A**, **65A** of the shroud walls **72**, **61** as shown in FIG. **6**. A radial outer surface of the body segment **84** is exposed to air that is radially between the carrier segments **32**, **38** and the blade track segments **34**, **40**.

(42) The first pocket **80** of the first blade track segment **34** and the second pocket **69** of the second blade track segment **40** retain the body segment **84** of the first strip seal **44** circumferentially between the first blade track segment **34** and the second blade track segment **40** as suggested in FIGS. **6** and **7**. The body segment **84** may move circumferentially a marginal amount, however, the pockets **80**, **69** block the body segment **84** from moving such that the circumferential gap **G** is no longer blocked.

(43) The second strip seal **46** is located radially inward of the first strip seal **44** as shown in FIG. **3**. The second strip seal **46** extends circumferentially into each of the first recess **78** of the first blade track segment **34** and the second recess **67** of the second blade track segment **40** to block the circumferential gap **G**. Because the second strip seal **46** is located radially inward of the first strip seal **44**, the second strip seal **46** provides a heat shield for the first strip seal **44** to protect the first strip seal **44** from heat of the gases in the gas path **25**.

(44) The second strip seal **46** includes an axial segment **90** and a forward radial segment **92** as shown in FIG. **3**. The axial segment **90** extends axially between a first end **90A** and a second end **90B** thereof opposite the first end **90A**. The forward radial segment **92** is coupled with the first end **90A** of the axial segment **90** to extend axially forward and radially outward from the first end **90A** of the axial segment **90** toward the first strip seal **44**. Illustratively, the forward radial segment **92** and the axial segment **90** both extend along straight paths, and an obtuse angle is formed between the forward radial segment **92** and the axial segment **90**. The second end **90B** of the axial segment **90** terminates in an axial direction.

(45) The axial segment **90** of the second strip seal **46** is located in the axially-extending portions **78B**, **67B** of the recesses **78**, **67** as shown in FIGS. **3** and **6**. The forward radial segment **92** is located in the first radially-extending portions **78A**, **67A** of the recesses **78**, **67** as shown in FIG. **7**.

(46) The forward radial segment **92** of the second strip seal **46**, the first strip seal **44**, and the axial segment **90** of the second strip seal **46** cooperate to form a cavity **94** radially between the first strip seal **44** and the second strip seal **46** as shown in FIGS. **3** and **6**. The cavity **94** is formed to include an at least partially closed axial forward end **94A** defined by the forward radial segment **92** of the second strip seal **46** and an open axial aft end **94B**. The open axial aft end **94B** of the cavity **94** is unobstructed such that the gases from the gas path **25** are free to flow into the cavity **94** through the open axial aft end **94B** thereof.

(47) As shown in FIG. **3**, the gases in the gas path **25** near the axial forward end **34A** of the first blade track segment **34** have a first pressure **P1** and the gases near an axial aft end **34C** of the first blade track segment **34** have a second pressure **P2** that is less than the first pressure **P1** caused by work being extracted from the gases by the blades **21**. Gases radially outward of the first strip seal **44** have a third pressure **P3** that is greater than the first pressure **P1** and the second pressure **P2**. In the illustrative embodiment, the third pressure **P3** is provided by compressor air (for example, compressor discharge air) that is conducted into the shroud segment **26**. The third pressure **P3** urges the first strip seal **44** radially inward. Due to the open axial aft end **94B** of the cavity **94**, gases from the gas path **25** may flow into the cavity **94** through the open axial aft end **94B**. Gases from the shroud segment **26** at the third pressure **P3** may leak into the cavity **94** between the first strip seal **44** and the second strip seal **46**. The gases in the cavity **94** have a fourth pressure **P4**. The fourth pressure **P4** is greater than the second pressure **P2** and less than the first pressure **P1** and the third pressure **P3**.

(48) Due to the third pressure **P3** of the gases radially outward of the first strip seal **44** being greater

than the fourth pressure **P4** of the gases in the cavity **94**, the first strip seal **44** is forced radially inwardly against the blade track segments **34**, **40**. The second strip seal **46**, thus, reduces a pressure load applied to the first strip seal **44** from the gases in the gas path **25** (as compared to direct exposure to the first pressure **P1**). As an example, without the second strip seal **46**, the gases in the gas path **25** may force the first strip seal **44** radially outward such that the first strip seal **44** does not engage each of the blade track segments **34**, **40**. The first pressure **P1** of the gases near the axial forward end **34A** of the first blade track segment **34** force the second strip seal **46** radially outwardly within the recesses **78**, **67** such that the axial segment **90** of the second strip seal **46** engages radially inwardly facing surfaces of the recesses **78**, **67** as shown in FIG. 6.

(49) In some embodiments, because the first strip seal **44** is forced radially inwardly due to the pressure differential between the third pressure **P3** and the fourth pressure **P4**, the damping segment **48** is omitted as shown in FIG. 8. In some embodiments, the damping segment **48** is included in the plurality of seals **30** as shown in FIG. 3. The damping segment **48** extends along a curvilinear path as shown in FIG. 3. In the illustrative embodiment, the curvilinear path forms a w-shape. The damping segment **48** is defined by a first radially-extending portion **96**, a second radially-extending portion **98**, and a curved intermediate portion **99** that extends between and interconnects the first radially-extending portion **96** and the second radially-extending portion **98**. The first radially-extending portion **96** forms the forward-most end of the damping segment **48**, and the second radially-extending portion **98** forms the aft-most end of the damping segment **48**.

(50) The curved intermediate portion **99** extends between a forward end **99A** and an aft end **99B** thereof. The first radially-extending portion **96** extends axially forward and radially outward from the forward end **99A** of the curved intermediate portion **99** and into the second slot **68** formed in the third flange **56** of the first carrier segment **32** as shown in FIG. 3. The first radially-extending portion **96** extends into both of the second slot **68** of the first carrier segment **32** and the fifth slot **57** of the second carrier segment **38**. The second radially-extending portion **98** extends axially aft and radially outward from the aft end **99B** of the curved intermediate portion **99** and into the third slot **70** formed in the fourth flange **58** of the first carrier segment **32** and the sixth slot **59** of the second carrier segment **38**. The curved intermediate portion **99**, from the forward end **99A** thereof, extends radially outward and axially aft to a peak **99C** as shown in FIG. 3. From the peak **99C**, the curved intermediate portion **99** extends radially inward and axially aft to the aft end **99B** thereof. The peak **99C** is located axially between the third flange **56** and the fourth flange **58**. The forward end **99A** and the aft end **99B** of the curved intermediate portion **99** each engage a radial outer surface of the body segment **84** of the first strip seal **44** as shown in FIG. 3. In some embodiments, the damping segment **48** is formed as a coil. In some embodiments, the damping segment **48** is u-shaped.

(51) The engagement between the first radially-extending portion **96** and the second slot **68** and the second radially-extending portion **98** and the third slot **70** applies a force to the body segment **84** of the first strip seal **44**. The force urges the body segment **84** of the first strip seal **44** radially inward against the first and second portions **76A**, **65A** of the shroud walls **72**, **61**.

(52) In some embodiments, the body segment **84** of the first strip seal **44** is formed to include at least one hole **27** as shown in FIG. 10 and as suggested in FIG. 9. The at least one hole **27** extends radially through the body segment **84** of the first strip seal **44** to direct cooling air **29** radially inwardly through the at least one hole **27** toward the second strip seal **46** to cool the second strip seal **46**. In some embodiments, the body segment **84** is formed to include a plurality of holes **27** that are axially spaced apart from one another.

(53) In some embodiments, the turbine shroud assembly **22** further includes strip seals **102**, **104**, **106**, **108**, **110** as shown in FIGS. 3 and 5. Each of the strip seals **102**, **104**, **106**, **108** extends into the first carrier segment **32** and the second carrier segment **38**. The strip seal **110** extends into each of the second attachment flange **74B** of the first blade track segment **34** and the fourth attachment flange **63B** of the second blade track segment **40**. The first carrier segment **32** and the second

carrier segment **38** are each formed to include grooves sized to receive the strip seals **102, 104, 106, 108** therein as shown in FIGS. **3** and **5**. The second attachment flange **74B** of the first blade track segment **34** and the fourth attachment flange **63B** of the second blade track segment **40** are each formed to include a groove sized to receive the strip seal **110** therein. The strip seals **102, 104, 106, 108, 110** provide additional sealing between the first shroud segment **26** and the second shroud segment **28**.

(54) Another embodiment of a turbine shroud assembly **222** in accordance with the present disclosure is shown in FIG. **11**. The turbine shroud assembly **222** is substantially similar to the turbine shroud assembly **22** shown in FIGS. **1-10** and described herein. Accordingly, similar reference numbers in the **200** series indicate features that are common between the turbine shroud assembly **22** and the turbine shroud assembly **222**. The description of the turbine shroud assembly **22** is incorporated by reference to apply to the turbine shroud assembly **222**, except in instances when it conflicts with the specific description and the drawings of the turbine shroud assembly **222**.

(55) As compared to the turbine shroud assembly **22**, the turbine shroud assembly **222** includes different shaped recesses **278, 267** formed in the blade track segments **234, 240** and a different shaped second strip seal **246**. The first recess **278** extends circumferentially into a first shroud wall **272** of the first blade track segment **234** to receive the second strip seal **246** therein as shown in FIG. **11**. The first recess **278** includes a first radially-extending portion **278A** that extends radially inward and axially aft and an axially-extending portion **278B** that extends axially aft from the first radially-extending portion **278A**. In some embodiments, the first radially-extending portion **278A** extends radially inward from a first portion **276A** of a first radial outer surface **276** of the first shroud wall **272**.

(56) A width of the axially-extending portion **278B** of the first recess **278** remains constant throughout the axially-extending portion **278B**. A width of the first radially-extending portion **278A** decreases as the first radially-extending portion **278A** extends radially outwardly from the axially-extending portion **278B** as shown in FIG. **11**. In other words, the first radially-extending portion **278A** is tapered as the first radially-extending portion **278A** extends radially outwardly from the axially-extending portion **278B**.

(57) The second recess **267** extends circumferentially into a second shroud wall **261** of the second blade track segment **240** to receive the second strip seal **246** therein as shown in FIG. **11**. The second recess **267** includes a first radially-extending portion **267A** that extends radially inward and axially aft and an axially-extending portion **267B** that extends axially aft from the first radially-extending portion **267A**. In some embodiments, the first radially-extending portion **267A** extends radially inward from a first portion **265A** of a second radial outer surface **265** of the second shroud wall **261**.

(58) A width of the axially-extending portion **267B** of the second recess **267** remains constant throughout the axially-extending portion **267B**. A width of the first radially-extending portion **267A** decreases as the first radially-extending portion **267A** extends radially outwardly from the axially-extending portion **267B** as shown in FIG. **11**. In other words, the first radially-extending portion **267A** is tapered as the first radially-extending portion **267A** extends radially outwardly from the axially-extending portion **267B**.

(59) As compared to the second strip seal **46**, the second strip seal **246** has a different shaped forward radial segment **292** as shown in FIG. **11**. An axial segment of the second strip seal **246** has a constant width throughout the axial segment. The forward radial segment **292** extends axially forward and radially outward from the axial segment toward the first strip seal **44**. A width of the forward radial segment **292** decreases as the forward radial segment **292** extends radially outwardly from the axial segment as shown in FIG. **11**. In other words, the forward radial segment **292** is tapered as the forward radial segment **292** extends radially outwardly. The tapered recesses **278, 267** and the tapered forward radial segment **292** reduce leakage from the gas path **25** radially outward through the recesses **278, 267**.

(60) Another embodiment of a turbine shroud assembly **322** in accordance with the present disclosure is shown in FIG. **12**. The turbine shroud assembly **322** is substantially similar to the turbine shroud assembly **22** shown in FIGS. **1-10** and described herein. Accordingly, similar reference numbers in the **300** series indicate features that are common between the turbine shroud assembly **22** and the turbine shroud assembly **322**. The description of the turbine shroud assembly **22** is incorporated by reference to apply to the turbine shroud assembly **322**, except in instances when it conflicts with the specific description and the drawings of the turbine shroud assembly **322**.

(61) As compared to the turbine shroud assembly **22**, the turbine shroud assembly **322** includes a different second strip seal **346** and a different recess **378**. The second strip seal **346** includes an axial segment **390** and an aft radial segment **393** as shown in FIG. **12**. The axial segment **390** extends axially between a first end **390A** and a second end **390B** thereof opposite the first end **390A**. The aft radial segment **393** is coupled with the second end **390B** of the axial segment **390** to extend axially aft and radially outward from the second end **390B** of the axial segment **390** toward the first strip seal **44**. Illustratively, the aft radial segment **393** and the axial segment **390** both extend along straight paths, and an obtuse angle is formed between the aft radial segment **393** and the axial segment **390**.

(62) The first recess **378** of the first shroud wall **72** includes an axially-extending portion **378B** that extends axially aft and a second radially-extending portion **378C** that extends radially outward and axially aft from the axially-extending portion **378B** as shown in FIG. **12**. In some embodiments, the second radially-extending portion **378C** extends radially outward to the first portion **76A** of the first radial outer surface **76** of the first shroud wall **72** such that the second radially-extending portion **378C** has an open end at the first portion **76A**. The axially-extending portion **378B** is radially spaced apart from the first portion **76A** of the first radial outer surface **76** of the first shroud wall **72**. The second recess of the second shroud wall of the second blade track segment is shaped the same as the first recess **378**.

(63) The aft radial segment **393** is located in the second radially-extending portions **378C** of the recesses **78** as shown in FIG. **12**. The first end **390A** of the axial segment **390** terminates in an axial direction. In some embodiments, the aft radial segment **393** of the second strip seal **346** and the second radially-extending portions **378C** of the recesses **378** may be tapered as described in relation to FIG. **11**.

(64) The aft radial segment **393** of the second strip seal **346**, the first strip seal **44**, and the axial segment **390** of the second strip seal **346** cooperate to form a cavity **394** radially between the first strip seal **44** and the second strip seal **346** as shown in FIG. **12**. The cavity **394** is formed to include an open axial forward end **394A** and an at least partially closed axial aft end **394B** defined by the aft radial segment **393** of the second strip seal **346**.

(65) Another embodiment of a turbine shroud assembly **422** in accordance with the present disclosure is shown in FIG. **13**. The turbine shroud assembly **422** is substantially similar to the turbine shroud assembly **22** shown in FIGS. **1-10** and described herein. Accordingly, similar reference numbers in the **400** series indicate features that are common between the turbine shroud assembly **22** and the turbine shroud assembly **422**. The description of the turbine shroud assembly **22** is incorporated by reference to apply to the turbine shroud assembly **422**, except in instances when it conflicts with the specific description and the drawings of the turbine shroud assembly **422**.

(66) As compared to the turbine shroud assembly **22**, the turbine shroud assembly **422** includes a different second strip seal **446** and a different recess **478**. The second strip seal **446** includes an axial segment **490**, a forward radial segment **492**, and an aft radial segment **493** as shown in FIG. **13**. The axial segment **490** extends axially between a first end **490A** and a second end **490B** thereof opposite the first end **490A**.

(67) The forward radial segment **492** is coupled with the first end **490A** of the axial segment **490** to extend axially forward and radially outward from the first end **490A** of the axial segment **490** toward the first strip seal **44**. Illustratively, the forward radial segment **492** and the axial segment

490 both extend along straight paths, and an obtuse angle is formed between the forward radial segment **492** and the axial segment **490**.

(68) The first recess **478** of the first shroud wall **72** includes a first radially-extending portion **478A** that extends radially inward and axially aft, an axially-extending portion **478B** that extends axially aft from the first radially-extending portion **478A**, and a second radially-extending portion **478C** that extends radially outward and axially aft from the axially-extending portion **478B** as shown in FIG. **13**. In some embodiments, the first radially-extending portion **478A** extends radially inward from the first portion **76A** of the first radial outer surface **76** of the first shroud wall **72** such that the first radially-extending portion **478A** has an open end at the first portion **76A**. In some embodiments, the second radially-extending portion **478C** extends radially outward to the first portion **76A** of the first radial outer surface **76** of the first shroud wall **72** such that the second radially-extending portion **478C** has an open end at the first portion **76A**. The axially-extending portion **478B** extends between and interconnects the first radially-extending portion **478A** and the second radially-extending portion **478C**. The axially-extending portion **478B** is radially spaced apart from the first portion **76A** of the first radial outer surface **76** of the first shroud wall **72**. The second recess of the second shroud wall of the second blade track segment is shaped the same as the first recess **478**.

(69) The forward radial segment **492** is located in the first radially-extending portions **478A** of the recesses **478** as shown in FIG. **13**. The aft radial segment **493** is coupled with the second end **490B** of the axial segment **490** to extend axially aft and radially outward from the second end **490B** of the axial segment **490** toward the first strip seal **44**. Illustratively, the aft radial segment **493** and the axial segment **490** both extend along straight paths, and an obtuse angle is formed between the aft radial segment **493** and the axial segment **490**. The aft radial segment **493** is located in the second radially-extending portions **478C** of the recesses **478** as shown in FIG. **13**. In some embodiments, the forward radial segment **492**, the aft radial segment **493**, the first radially-extending portions **478A** of the recesses **78**, and the second radially-extending portions **478C** of the recesses **78** may be tapered as described in relation to FIG. **11**.

(70) The aft radial segment **493** of the second strip seal **446**, the first strip seal **44**, the forward radial segment **492** of the second strip seal **446**, and the axial segment **490** of the second strip seal **446** cooperate to form a cavity **494** radially between the first strip seal **44** and the second strip seal **446** as shown in FIG. **13**. The cavity **494** is formed to include an at least partially closed axial forward end **494A** defined by the forward radial segment **492** of the second strip seal **446** and an at least partially closed axial aft end **494B** defined by the aft radial segment **493** of the second strip seal **446**.

(71) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A turbine shroud assembly for use with a gas turbine engine, the turbine shroud assembly comprising: a first shroud segment including a first carrier segment arranged circumferentially at least partway around a central axis and a first blade track segment supported by the first carrier segment to define a first portion of a gas path of the turbine shroud assembly, the first blade track segment having a first shroud wall that extends circumferentially partway around the central axis, a first attachment flange that extends radially outward from the first shroud wall, and a second attachment flange that extends radially outward from the first shroud wall axially spaced apart from the first attachment flange, wherein the first shroud wall has a first radial outer surface and a first radial inner surface, the first radial outer surface includes a first portion and a second portion that

extends circumferentially away from the first portion and is spaced radially outward from the first portion, and the first shroud wall is formed to include a first recess that extends circumferentially into the first shroud wall, a second shroud segment arranged circumferentially adjacent the first shroud segment about the central axis, the second shroud segment including a second carrier segment arranged circumferentially at least partway around the central axis and a second blade track segment supported by the second carrier segment to define a second portion of the gas path of the turbine shroud assembly, the second blade track segment having a second shroud wall that extends circumferentially partway around the central axis, a first attachment flange that extends radially outward from the second shroud wall, and a second attachment flange that extends radially outward from the second shroud wall axially spaced apart from the first attachment flange of the second blade track segment, wherein the second shroud wall has a second radial outer surface and a second radial inner surface, the second radial outer surface includes a first portion and a second portion that extends circumferentially away from the first portion of the second radial outer surface and is spaced radially outward from the first portion of the second radial outer surface, and the second shroud wall is formed to include a second recess that extends circumferentially into the second shroud wall, and a plurality of seals extending circumferentially into the first shroud segment and the second shroud segment to block gases from escaping the gas path radially between the first shroud segment and the second shroud segment, the plurality of seals including a first strip seal and a second strip seal, the first strip seal extends axially along the first portion of the first radial outer surface of the first shroud wall and the first portion of the second radial outer surface of the second shroud wall to block the gases from passing radially between and beyond the first shroud wall and the second shroud wall, and the second strip seal is located radially inward of the first strip seal and extends circumferentially into the first recess formed in the first shroud wall of the first blade track segment and the second recess formed in the second shroud wall of the second blade track segment to provide a heat shield for the first strip seal to protect the first strip seal from heat of the gases in the gas path and to reduce a pressure load applied to the first strip seal from the gases in the gas path, wherein the first strip seal is formed to include at least one hole extending radially through the first strip seal to direct cooling air radially inward through the at least one hole toward the second strip seal to cool the second strip seal.

2. The turbine shroud assembly of claim 1, wherein the second strip seal includes an axial segment that extends axially between a first end and a second end thereof opposite the first end.

3. The turbine shroud assembly of claim 2, wherein the second strip seal further includes a forward radial segment coupled with the first end of the axial segment and extending axially forward and radially outward from the first end of the axial segment toward the first strip seal.

4. The turbine shroud assembly of claim 3, wherein the second end of the axial segment terminates in an axial direction to cause the forward radial segment and the axial segment to cooperate with the first strip seal to define a cavity radially between the first strip seal and the axial segment of the second strip seal, the cavity having an at least partially closed axial forward end defined by the forward radial segment and an open axial aft end that allows a first portion of the gases to flow into the cavity, the first portion of the gases in the cavity having a relatively lower pressure than a second portion of the gases in the gas path that have a relatively higher pressure so that the second portion of the gases urge the second strip seal radially outward within the first recess and the second recess.

5. The turbine shroud assembly of claim 3, wherein the axial segment of the second strip seal has a constant width between the first end and the second end thereof and the forward radial segment of the second strip seal has a tapered width that decreases as the forward radial segment extends radially outward from the first end of the axial segment toward the first strip seal.

6. The turbine shroud assembly of claim 3, wherein the first recess of the first shroud wall includes a radially-extending portion that extends radially inward and axially aft and an axially-extending portion that extends axially aft from the first radially-extending portion.

7. The turbine shroud assembly of claim 6, wherein the radially-extending portion extends radially inward from the first portion of the first radial outer surface of the first shroud wall, and the axially-extending portion is radially spaced apart from the first portion of the first radial outer surface of the first shroud wall.

8. The turbine shroud assembly of claim 6, wherein the axial segment of the second strip seal extends circumferentially into the axially-extending portion of the first recess and the forward radial segment of the second strip seal extends circumferentially into the radially-extending portion of the first recess.

9. The turbine shroud assembly of claim 1, wherein the first recess of the first shroud wall includes a radially-extending portion that extends radially inward and axially aft and an axially-extending portion that extends axially aft from the radially-extending portion.

10. The turbine shroud assembly of claim 1, wherein the first carrier segment includes a first outer wall, a first flange that extends radially inward from the first outer wall, and a second flange axially spaced apart from the first flange and extending radially inward from the first outer wall, and wherein the first strip seal includes a body segment that extends axially along the first portion of the first radial outer surface of the first shroud wall and the first portion of the second radial outer surface of the second shroud wall, a forward segment coupled to a first end of the body segment and extending axially forward and radially outward from the first end of the body segment into the first flange of the first carrier segment, and an aft segment coupled to a second end of the body segment opposite the first end thereof and extending axially aft and radially outward from the second end of the body segment, and wherein the aft segment is located axially aft of the second flange of the first carrier segment.

11. A turbine shroud assembly for use with a gas turbine engine, the turbine shroud assembly comprising: a first shroud segment including a first carrier segment arranged circumferentially at least partway around a central axis and a first blade track segment supported by the first carrier segment to define a first portion of a gas path of the turbine shroud assembly, the first blade track segment having a first shroud wall that extends circumferentially partway around the central axis and a first attachment feature that extends radially outward from the first shroud wall, the first shroud wall defining a first radial outer surface, a second shroud segment arranged circumferentially adjacent the first shroud segment about the central axis, the second shroud segment including a second carrier segment arranged circumferentially at least partway around the central axis and a second blade track segment supported by the second carrier segment to define a second portion of the gas path of the turbine shroud assembly, the second blade track segment having a second shroud wall that extends circumferentially partway around the central axis and a second attachment feature that extends radially outward from the second shroud wall, the second shroud wall defining a second radial outer surface, and a plurality of seals extending circumferentially into the first shroud segment and the second shroud segment, the plurality of seals including a first seal and a second seal, the first seal extends axially along the first radial outer surface of the first shroud wall and the second radial outer surface of the second shroud wall, and the second seal is located radially inward of the first seal and extends circumferentially into the first blade track segment and the second blade track segment, wherein the second seal includes an axial segment that extends axially between a first end and a second end thereof opposite the first end and a forward radial segment coupled with the first end of the axial segment and extending axially forward and radially outward from the first end of the axial segment toward the first seal.

12. The turbine shroud assembly of claim 11, wherein the second end of the axial segment terminates in an axial direction to cause the forward radial segment and the axial segment to cooperate with the first seal to define a cavity radially between the first seal and the axial segment of the second seal, the cavity having an at least partially closed axial forward end defined by the forward radial segment and an open axial aft end.

13. The turbine shroud assembly of claim 11, wherein the axial segment of the second seal has a

constant width between the first end and the second end thereof and the forward radial segment of the second seal has a tapered width that decreases as the forward radial segment extends radially outward from the first end of the axial segment toward the first seal.

14. The turbine shroud assembly of claim 11, wherein the first shroud wall is formed to include a first recess that extends circumferentially into the first shroud wall and the second shroud wall is formed to include a second recess that extends circumferentially into the second shroud wall, and wherein the second seal extends circumferentially into each of the first recess and the second recess to extend therebetween.

15. The turbine shroud assembly of claim 14, wherein the first recess of the first shroud wall includes a radially-extending portion that extends radially inward and axially aft and an axially-extending portion that extends axially aft from the radially-extending portion.

16. The turbine shroud assembly of claim 11, wherein the first carrier segment includes a first outer wall, a first flange that extends radially inward from the first outer wall, and a second flange axially spaced apart from the first flange and extending radially inward from the first outer wall, and wherein the first seal includes a body segment that extends axially along the first radial outer surface of the first shroud wall and the second radial outer surface of the second shroud wall, a forward segment coupled to a first end of the body segment and extending axially forward and radially outward from the first end of the body segment into the first flange of the first carrier segment, and an aft segment coupled to a second end of the body segment opposite the first end thereof and extending axially aft and radially outward from the second end of the body segment, and wherein the aft segment is located axially aft of the second flange of the first carrier segment.

17. The turbine shroud assembly of claim 11, wherein the first seal is formed to include at least one hole extending radially through the first seal to direct cooling air radially inward through the at least one hole toward the second seal to cool the second seal.

18. A method comprising: assembling a first shroud segment by coupling a first blade track segment with a first carrier segment to support the first blade track segment radially inward of the first carrier segment, assembling a second shroud segment by coupling a second blade track segment with a second carrier segment to support the second blade track segment radially inward of the second carrier segment, locating a first seal on a first radial outer surface of the first blade track segment and a second radial outer surface of the second blade track segment, locating a second seal in a first recess that extends circumferentially into the first blade track segment and in a second recess that extends circumferentially into the second blade track segment so that the second seal is located radially inwardly of the first seal, and shielding the first seal from heat of gases in a gas path via the second seal, wherein the second seal includes an axial segment that extends axially between a first end and a second end thereof opposite the first end and a forward radial segment coupled with the first end of the axial segment and extending axially forward and radially outward from the first end of the axial segment toward the first seal.

19. The method of claim 18, wherein the second end of the axial segment terminates in an axial direction to cause the forward radial segment and the axial segment to cooperate with the first seal to define a cavity radially therebetween, and wherein the method further comprises directing a first portion of the gases having a first pressure into the cavity at the second end of the axial segment and urging the second seal radially outward within the first recess and the second recess via a second portion of the gases flowing radially inward of the second seal having a second pressure that is greater than the first pressure.

20. The method of claim 19, further comprising urging the first seal radially inward against the first blade track segment and the second blade track segment via a third portion of the gases flowing radially outward of the first seal having a third pressure that is greater than the first pressure.
